

ER Diagram Design and Relational Schema

E-Commerce Order Management Database System

Field	Details
Course	Database Management Systems (Skill Development Activity)
Task	Task 3: Design the ER Diagram and convert it into a Relational Schema
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Submitted to	[Faculty Name]
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1. Introduction

In Task 2, nine entities, their attributes, keys and relationships were identified. This task presents that design visually as an Entity Relationship (ER) diagram and then converts the ER model into a relational schema, which is the set of tables that will be created in SQL in the implementation phase.

1.1 Objectives

1. Draw a conceptual ER diagram showing entities, relationships and cardinalities.
2. Draw a logical ER diagram showing every table with its attributes, primary keys and foreign keys.
3. Convert the ER model into a relational schema using standard mapping rules.
4. Define referential integrity rules (foreign key actions) between the tables.

1.2 ER Notation Used

Symbol	Meaning
Rectangle	Entity (a table in the relational model)
Diamond	Relationship between entities (conceptual diagram)
Dashed oval	Attribute of a relationship (for example quantity of an order line)
1, N, M	Cardinality: one, or many, at that end of the relationship
Two bars (logical diagram)	Exactly one record at that end
Crow's foot (logical diagram)	Many records at that end
PK / FK / UK	Primary key / Foreign key / Unique key

2. Entities and Relationships at a Glance

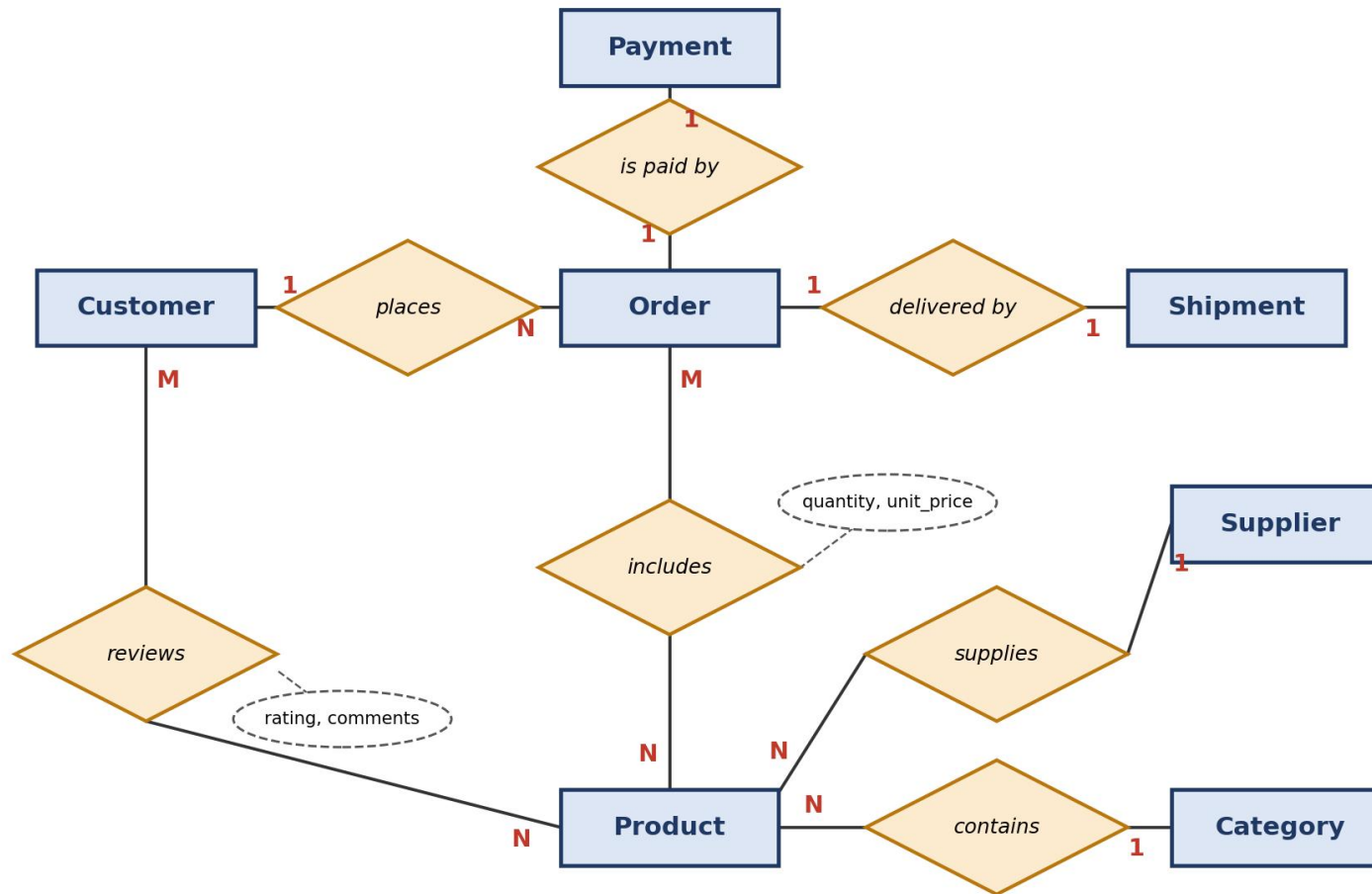
Entity	Role in the system
Customer	Registered buyer who places orders and writes reviews
Category	Classification group for products
Supplier	Vendor who supplies products
Product	Item for sale with price and stock

Entity	Role in the system
Order	Purchase made by a customer (table name ORDERS because ORDER is an SQL keyword)
Order Details	Associative entity linking Order and Product, carrying quantity and unit price
Payment	Payment made for an order
Shipment	Delivery of an order
Review	Associative entity linking Customer and Product, carrying rating and comments

Relationship	Cardinality
Customer places Order	One-to-Many (1:N)
Category contains Product	One-to-Many (1:N)
Supplier supplies Product	One-to-Many (1:N)
Order includes Product	Many-to-Many (M:N), resolved by Order Details
Customer reviews Product	Many-to-Many (M:N), resolved by Review
Order is paid by Payment	One-to-One (1:1)
Order is delivered by Shipment	One-to-One (1:1)

3. Conceptual ER Diagram

The conceptual diagram shows the main entities and how they are related. The many-to-many relationships (Order–Product and Customer–Product) are shown directly with the attributes that belong to the relationship itself.



Conceptual ER diagram | 1 = one, N / M = many | dashed ovals = attributes of the relationship

Figure 1: Conceptual ER diagram of the E-Commerce Order Management System

4. Logical ER Diagram

In the logical diagram, each many-to-many relationship is replaced by an associative table (ORDER_DETAILS and REVIEW), foreign keys are shown explicitly, and every attribute has a data type.

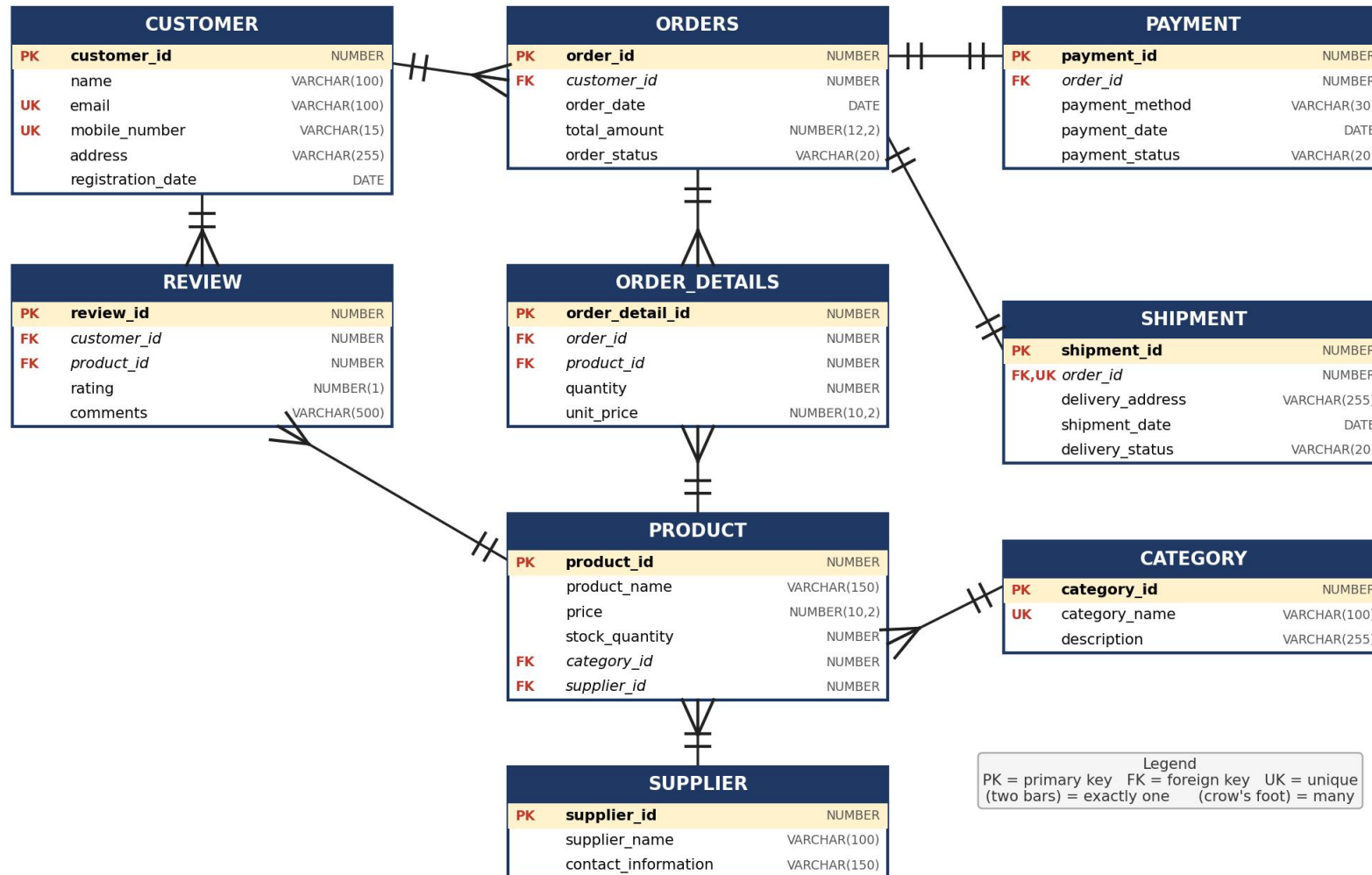


Figure 2: Logical ER diagram with attributes, primary keys and foreign keys

5. Conversion of ER Model to Relational Schema

5.1 Mapping Rules Applied

Rule	ER construct	Conversion to relational model
1	Strong entity	Becomes a table. Simple attributes become columns. The key attribute becomes the primary key.
2	One-to-Many (1:N) relationship	The primary key of the 'one' side is added as a foreign key in the table on the 'many' side.
3	One-to-One (1:1) relationship	The primary key of one table is added as a foreign key in the other, with a UNIQUE constraint so that only one record can match.
4	Many-to-Many (M:N) relationship	A new associative table is created. It holds the primary keys of both entities as foreign keys, plus any attributes of the relationship.
5	Relationship attributes	Attributes of a relationship (quantity, unit price, rating, comments) become columns of the associative table.
6	Candidate keys	Alternate keys such as email and mobile number are declared UNIQUE.

5.2 Application of the Rules to Each Relationship

Relationship	Rule	Result in the schema
Customer places Order (1:N)	2	ORDERS.customer_id is a foreign key to CUSTOMER
Category contains Product (1:N)	2	PRODUCT.category_id is a foreign key to CATEGORY
Supplier supplies Product (1:N)	2	PRODUCT.supplier_id is a foreign key to SUPPLIER
Order includes Product (M:N)	4, 5	New table ORDER_DETAILS(order_id, product_id, quantity, unit_price)
Customer reviews Product (M:N)	4, 5	New table REVIEW(customer_id, product_id, rating, comments)
Order is paid by Payment (1:1)	3	PAYMENT.order_id is a foreign key to ORDERS
Order is delivered by Shipment (1:1)	3	SHIPMENT.order_id is a foreign key to ORDERS with a UNIQUE constraint

6. Relational Schema

6.1 Schema in Notation Form

Primary keys are underlined in textbooks; here they are marked (PK). Foreign keys are marked (FK) with the referenced table.

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CUSTOMER (customer_id (PK), name, email, mobile_number, address, registration_date)
CATEGORY (category_id (PK), category_name, description)
SUPPLIER (supplier_id (PK), supplier_name, contact_information)
PRODUCT (product_id (PK), product_name, price, stock_quantity, category_id (FK to CATEGORY),
supplier_id (FK to SUPPLIER))
ORDERS (order_id (PK), customer_id (FK to CUSTOMER), order_date, total_amount, order_status)
ORDER_DETAILS (order_detail_id (PK), order_id (FK to ORDERS), product_id (FK to PRODUCT),
quantity, unit_price)
PAYMENT (payment_id (PK), order_id (FK to ORDERS), payment_method, payment_date,
payment_status)
SHIPMENT (shipment_id (PK), order_id (FK to ORDERS), delivery_address, shipment_date,
delivery_status)
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REVIEW (review_id (PK), customer_id (FK to CUSTOMER), product_id (FK to PRODUCT), rating, comments)

6.2 Table-wise Schema with Data Types and Constraints

CUSTOMER

Column	Data type	Constraint
customer_id	NUMBER	PRIMARY KEY
name	VARCHAR(100)	NOT NULL
email	VARCHAR(100)	NOT NULL, UNIQUE
mobile_number	VARCHAR(15)	NOT NULL, UNIQUE
address	VARCHAR(255)	NOT NULL
registration_date	DATE	DEFAULT current date

CATEGORY

Column	Data type	Constraint
category_id	NUMBER	PRIMARY KEY
category_name	VARCHAR(100)	NOT NULL, UNIQUE
description	VARCHAR(255)	Optional

SUPPLIER

Column	Data type	Constraint
supplier_id	NUMBER	PRIMARY KEY
supplier_name	VARCHAR(100)	NOT NULL
contact_information	VARCHAR(150)	NOT NULL

PRODUCT

Column	Data type	Constraint
product_id	NUMBER	PRIMARY KEY
product_name	VARCHAR(150)	NOT NULL
price	NUMBER(10,2)	NOT NULL, CHECK (price > 0)
stock_quantity	NUMBER	NOT NULL, DEFAULT 0, CHECK (stock_quantity >= 0)
category_id	NUMBER	FOREIGN KEY references CATEGORY(category_id)
supplier_id	NUMBER	FOREIGN KEY references SUPPLIER(supplier_id)

ORDERS

Column	Data type	Constraint
order_id	NUMBER	PRIMARY KEY
customer_id	NUMBER	NOT NULL, FOREIGN KEY references CUSTOMER(customer_id)
order_date	DATE	DEFAULT current date
total_amount	NUMBER(12,2)	CHECK (total_amount >= 0)

Column	Data type	Constraint
order_status	VARCHAR(20)	CHECK in (Placed, Confirmed, Shipped, Delivered, Cancelled)

ORDER_DETAILS

Column	Data type	Constraint
order_detail_id	NUMBER	PRIMARY KEY
order_id	NUMBER	NOT NULL, FOREIGN KEY references ORDERS(order_id)
product_id	NUMBER	NOT NULL, FOREIGN KEY references PRODUCT(product_id)
quantity	NUMBER	NOT NULL, CHECK (quantity > 0)
unit_price	NUMBER(10,2)	NOT NULL, CHECK (unit_price > 0)
(order_id, product_id)	Composite	UNIQUE (a product appears once per order)

PAYMENT

Column	Data type	Constraint
payment_id	NUMBER	PRIMARY KEY
order_id	NUMBER	NOT NULL, FOREIGN KEY references ORDERS(order_id)
payment_method	VARCHAR(30)	CHECK in (UPI, Credit Card, Debit Card, Net Banking, Cash on Delivery)
payment_date	DATE	DEFAULT current date
payment_status	VARCHAR(20)	CHECK in (Pending, Completed, Failed, Refunded)

SHIPMENT

Column	Data type	Constraint
shipment_id	NUMBER	PRIMARY KEY
order_id	NUMBER	NOT NULL, UNIQUE, FOREIGN KEY references ORDERS(order_id)
delivery_address	VARCHAR(255)	NOT NULL
shipment_date	DATE	Optional until dispatched
delivery_status	VARCHAR(20)	CHECK in (Processing, Shipped, Out for Delivery, Delivered, Returned)

REVIEW

Column	Data type	Constraint
review_id	NUMBER	PRIMARY KEY
customer_id	NUMBER	NOT NULL, FOREIGN KEY references CUSTOMER(customer_id)
product_id	NUMBER	NOT NULL, FOREIGN KEY references PRODUCT(product_id)
rating	NUMBER(1)	NOT NULL, CHECK (rating BETWEEN 1 AND 5)
comments	VARCHAR(500)	Optional

Column	Data type	Constraint
(customer_id, product_id)	Composite	UNIQUE (one review per customer per product)

7. Referential Integrity (Foreign Key Rules)

Referential integrity ensures that a foreign key value always matches an existing primary key value in the parent table, or is NULL where allowed.

Child table (FK)	Parent table (PK)	On delete of parent	Reason
PRODUCT.category_id	CATEGORY	RESTRICT	A category with products cannot be deleted
PRODUCT.supplier_id	SUPPLIER	RESTRICT	A supplier with products cannot be deleted
ORDERS.customer_id	CUSTOMER	RESTRICT	Order history must be preserved
ORDER_DETAILS.order_id	ORDERS	CASCADE	Line items have no meaning without their order
ORDER_DETAILS.product_id	PRODUCT	RESTRICT	Sold products must remain in records
PAYMENT.order_id	ORDERS	RESTRICT	Financial records must be kept
SHIPMENT.order_id	ORDERS	RESTRICT	Delivery records must be kept
REVIEW.customer_id	CUSTOMER	CASCADE	Reviews can be removed with the customer
REVIEW.product_id	PRODUCT	CASCADE	Reviews can be removed with the product

8. Verification of the Schema

Check	Result
Every table has a primary key	Yes, all 9 tables
Every 1:N relationship has a foreign key on the many side	Yes (ORDERS, PRODUCT x2, ORDER_DETAILS x2, REVIEW x2)
Every 1:1 relationship has a UNIQUE foreign key	SHIPMENT.order_id is UNIQUE; PAYMENT.order_id can be made UNIQUE if only one payment per order is allowed
Every M:N relationship is resolved	Yes, through ORDER_DETAILS and REVIEW
No multivalued or composite attributes	Yes, all attributes are atomic, which supports First Normal Form
Total tables in the schema	9
Total foreign keys	9

9. Conclusion

The conceptual and logical ER diagrams represent the e-commerce system with nine entities connected by seven relationships. By applying the standard mapping rules, the ER model was converted into a relational schema of nine tables with primary keys, foreign keys, unique keys, check and default constraints, and defined foreign key delete rules. This schema is the starting point for the normalization work in the next phase and for the SQL table creation that follows.